

Deep Learning for Solving Dynamic Models

Session 7: Structural estimation via simulated method of moments

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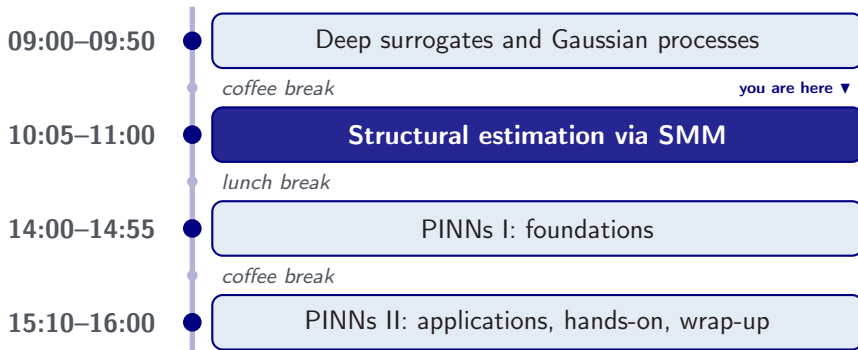
HEC, University of Lausanne

Based on Chen, Didisheim & Scheidegger (2026), *Journal of Financial Economics* 177, 104222

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Day 2 at a Glance



Roadmap of This Lecture (55 min)

I. The idea (12 min)

- ▶ The estimation loop, and why it is expensive
- ▶ SMM on one frame; common random numbers
- ▶ The model and the moments

II. One parameter (18 min)

- ▶ A policy network with ϱ as an input
- ▶ Its accuracy against the reference solution
- ▶ Moments, criterion, estimate (07_01)

Notebooks in day2/code/07_structural_estimation/: 07_01_SMM_One_Parameter.ipynb,
07_02_SMM_Two_Parameters.ipynb

III. Two parameters (15 min)

- ▶ (β, ϱ) jointly (07_02)
- ▶ The valley and the basin: which moments identify
- ▶ The moment Jacobian in one number

Finger exercise, production, hands-on (10 min)

Learning Objectives

By the end of this lecture you should be able to:

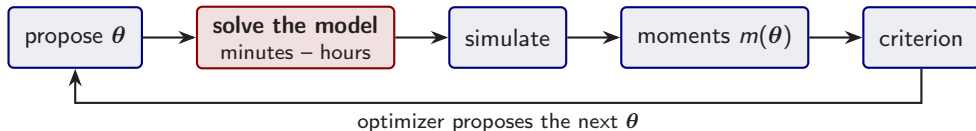
- ▶ Write down the simulated-method-of-moments criterion and say why every evaluation of it used to require solving the model
- ▶ Train a policy network with structural parameters as inputs, and validate it against a reference solution across the parameter range
- ▶ Simulate moments under common random numbers and minimize the criterion on the fly, for one parameter and for two
- ▶ Read a criterion surface: tell an identified basin from a flat valley, and confirm it with the moment Jacobian

Part I

The Idea

The Estimation Loop, and Why It Is Expensive

The nested loop. For each candidate θ :



The surrogate move. Train the policy network with θ as an input, once. The red box is then gone from the loop: a candidate θ costs one simulation of the network, and the loop runs on the fly.

Two consequences

The criterion is cheap enough to evaluate on a grid, so you can *look at* it rather than trust a local optimizer's report; and under common random numbers it is smooth in θ , so a standard minimizer finishes the job.

SMM on One Frame

Data: T observations $\{y_t\}$, summarized by q moments $\hat{m} = \frac{1}{T} \sum_t g(y_t)$: here standard deviations, autocorrelations, a mean.

Model: for a candidate θ , simulate a sample of length S and compute the same moments, $m(\theta)$.

The estimator

$$\hat{\theta} = \arg \min_{\theta} Q(\theta), \quad Q(\theta) = [m(\theta) - \hat{m}]^T W [m(\theta) - \hat{m}], \quad W = I \text{ here.}$$

- ▶ $q \geq p$ moments for p parameters; with $q > p$ the criterion cannot reach zero and its minimum is a compromise across moments.
- ▶ $W = I$ is consistent and simple; the efficient choice, the inverse covariance of the data moments, changes the standard errors, not the picture.
- ▶ **Identification:** θ_0 is the unique minimizer of the population criterion, so the moments must *move* with every parameter (Part III).

Common random numbers

Use the **same** shock sequence for every candidate θ . Then $m(\theta)$ is a smooth function and Q has a clean minimum. Redraw the shocks and the minimizer chases simulation noise.

The Model: Brock–Mirman with Stochastic TFP

The economy of Session 2, with $\delta = 0.10$ so that no closed form exists and the Euler equation has to be solved:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[\ln C_t], \quad K_{t+1} + C_t = Y_t + (1 - \delta)K_t,$$

$$Y_t = z_t K_t^\alpha, \quad \log z_{t+1} = \varrho \log z_t + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1).$$

Calibration

α	δ	σ	β (Part II)
0.36	0.10	0.04	0.96

Truth for the synthetic data:

$\varrho = 0.90$, and in Part III $\beta = 0.96$.

Moments, and what moves them

Std($\Delta \log C_t$)	persistence, weakly
Corr($\Delta \log C_t, \Delta \log C_{t-1}$)	persistence
Corr($\log Y_t, \log Y_{t-1}$)	persistence, directly
mean(I_t/Y_t)	the level of capital: β

ϱ sets how long shocks last, so autocorrelations identify it. β sets the steady-state capital stock, $k^* = (\alpha/(1/\beta - 1 + \delta))^{1/(1-\alpha)}$, so a level moment identifies it and the dynamic ones barely do.

The Reference Solution, and the Data

Reference solver (`brock_mirman_reference.py`): iterate on the Euler equation with cash on hand as the state (endogenous grid method, cubic splines) until the consumption function stops changing. Relative Euler errors below 10^{-6} on the ergodic set at every corner of the parameter box; about 1.7 s per solve.

The data: one path of $T = 500$ periods from the reference solution at the true parameters, with its own shocks. Its four moments are the targets $\hat{m}$.

What the reference solver is for

Not for the estimation, where it would be the expensive red box. For **validating the surrogate**, at several parameter values, on the states the model visits; and, once, for drawing the criterion the true model would give, so that the surrogate's criterion can be laid on top of it.

Part II

One Parameter: ϱ

Step 1: A Policy Network with ϱ as an Input

The surrogate

$(\log z_t, K_t, \varrho) \mapsto$ consumption share of cash on hand $m_t = Y_t + (1 - \delta)K_t$, through a sigmoid, so $0 < C_t < m_t$ by construction and $K_{t+1} = m_t - C_t$.

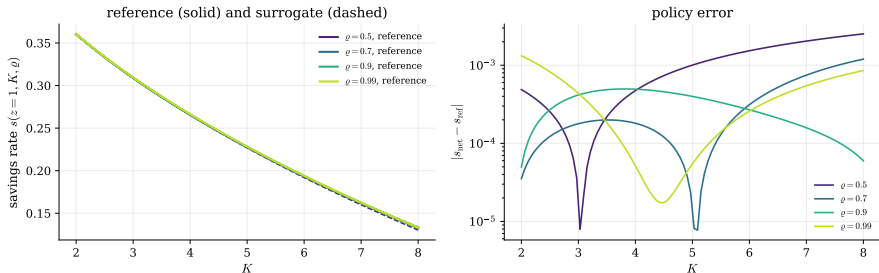
Loss: the relative Euler residual, with five Gauss–Hermite nodes for the expectation,

$$e(z, K, \varrho) = 1 - \frac{1}{\beta C_t \mathbb{E}_t \left[\frac{1 - \delta + \alpha z_{t+1} K_{t+1}^{\alpha-1}}{C_{t+1}} \right]}, \quad \mathcal{L} = \frac{1}{N} \sum_i e_i^2.$$

States: each batch is half uniform draws over a box in $(\log z, K, \varrho)$, half states visited by simulating the current policy, so the network is accurate where the estimator will evaluate it.

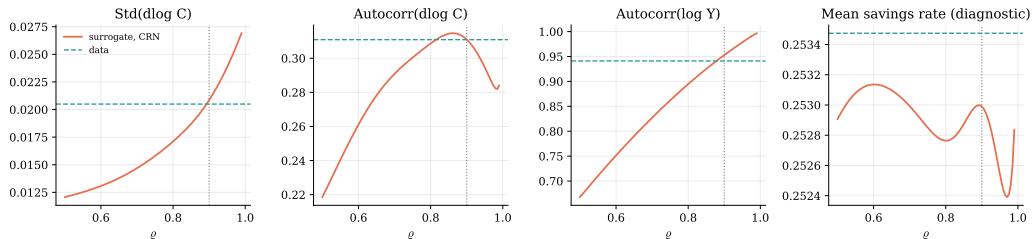
Nothing about ϱ is special. It enters the transition $\log z_{t+1} = \varrho \log z_t + \sigma \varepsilon$ inside the expectation, and the Euler residual is zero for every value of it. One training run, 2 min, covers $\varrho \in [0.50, 0.99]$.

Step 1, Checked Against the Reference Solution



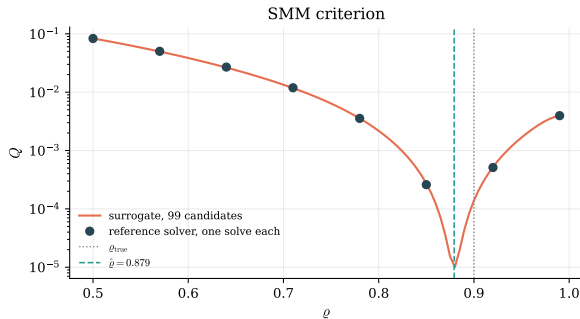
Notebook 07_01, stored teaching run. On 2,000 ergodic states per ϱ : $\varrho = 0.5$: mean Euler error $1.6 \cdot 10^{-4}$, mean consumption error $8.3 \cdot 10^{-4}$, max $1.3 \cdot 10^{-3}$; $\varrho = 0.7$: mean Euler error $5.5 \cdot 10^{-5}$, mean consumption error $2.1 \cdot 10^{-4}$, max $4.9 \cdot 10^{-4}$; $\varrho = 0.9$: mean Euler error $1.1 \cdot 10^{-4}$, mean consumption error $5.7 \cdot 10^{-4}$, max $7.1 \cdot 10^{-4}$; $\varrho = 0.99$: mean Euler error $2.5 \cdot 10^{-4}$, mean consumption error $1.1 \cdot 10^{-3}$, max $2.1 \cdot 10^{-2}$. The consumption error is the number that matters for estimation; the Euler residual is only the training loss.

Step 2: Moments Under Common Random Numbers



Notebook 07_01: 99 candidates from 0.50 to 0.99, each simulated for 2,000 periods from the same initial state under the same shocks, in 2 s. The dashed lines are the data moments. The output autocorrelation rises monotonically in ϱ and identifies it on its own; the consumption-growth autocorrelation rises up to $\varrho \approx 0.9$ and falls beyond it, so near the truth it adds curvature but not sign; the standard deviation rises steeply only above 0.8; the mean investment share is flat, as it should be, and is not used.

Step 3: The Criterion and the Estimate



Notebook 07_01. The surrogate's criterion on 99 candidates, and the reference solver's on 8 candidates, one solve each. Grid minimum, then a bounded scalar minimizer: $\hat{\theta} = 0.8793$ against 0.90. Two things keep the estimate off the truth, and the figure separates them: the 500-period sample (the reference criterion is not zero at 0.90 either) and the surrogate's error (the gap between the two curves).

Part III

Two Parameters: (β, ϱ)

The Same Surrogate, One More Input

$(\log z_t, K_t, \beta, \varrho) \mapsto$ consumption share; the Euler residual with β read from the input. Box $\beta \in [0.92, 0.99]$, $\varrho \in [0.50, 0.99]$; the capital box is wide enough for every β 's ergodic set.

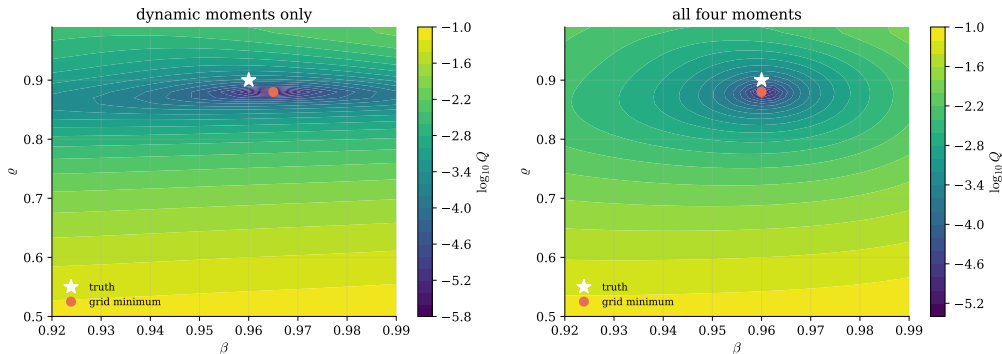
Checked at five corners and the centre of the box, on 2,000 ergodic states each:

β	ϱ	mean Euler	max Euler	mean $ \Delta C /C$	max $ \Delta C /C$
0.92	0.5	$1.3 \cdot 10^{-4}$	$3.5 \cdot 10^{-4}$	$4.2 \cdot 10^{-4}$	$7.6 \cdot 10^{-4}$
0.92	0.99	$1.8 \cdot 10^{-4}$	$1.1 \cdot 10^{-3}$	$5.9 \cdot 10^{-4}$	$3.0 \cdot 10^{-3}$
0.96	0.9	$3.1 \cdot 10^{-5}$	$1.9 \cdot 10^{-4}$	$9.8 \cdot 10^{-5}$	$5.3 \cdot 10^{-4}$
0.99	0.5	$3.0 \cdot 10^{-4}$	$4.4 \cdot 10^{-4}$	$2.3 \cdot 10^{-3}$	$2.9 \cdot 10^{-3}$
0.99	0.99	$5.3 \cdot 10^{-4}$	$5.1 \cdot 10^{-3}$	$3.7 \cdot 10^{-3}$	$3.3 \cdot 10^{-2}$

Why the check comes first

A criterion that is flat in β has two possible causes: the moments do not respond to β , or the *surrogate* does not. Only the comparison with the reference solution tells them apart. The table rules the second out here.

Two Criterion Surfaces: the Valley and the Basin



Notebook 07_02, 1,450 candidates under common random numbers. Left: the three dynamic moments only. A long valley along β : many pairs fit about equally well, and the minimizer's answer along it is arbitrary. Right: with the mean investment share added, the valley closes into a basin. **Identification is fixed by information, not by a better optimizer.**

The Moment Jacobian in One Number

At the estimate, by central differences on the surrogate, with the parameters scaled to the box, $u = (\theta - \theta_{lo})/(\theta_{hi} - \theta_{lo})$, so that a step of one is a step across the plotted range in either direction (in raw units the β column would be seven times smaller only because the box is seven times narrower):

$$M = \frac{\partial m}{\partial u} \in \mathbb{R}^{q \times 2}, \quad M = U \text{diag}(s_1, s_2) V^\top.$$

Near the minimum $Q \approx \delta u^\top M^\top M \delta u$, so s_2^2 is the criterion's curvature in its flattest direction and the last column of V names that direction.

moments	s_1	s_2	flattest direction (β, ϱ)
dynamic only	0.273	0.0320	$(-1.00, -0.01)$
all four	0.272	0.1419	$(-1.00, -0.02)$

Estimates: dynamic moments only $(\hat{\beta}, \hat{\varrho}) = (0.9641, 0.8786)$; all four moments $(0.9605, 0.8782)$; truth $(0.96, 0.90)$. The ratio of singular values falls from 9 to 2 when the level moment is added.

With an efficient W and the covariance of the data moments, $(M^\top W M)^{-1}$ scaled by $1/T$ is the asymptotic variance of $\hat{u}$; a small s_2 is a large standard error along V_2 . The notebooks stop at M ; the covariance is one more simulation away.

The Surrogate Is Not Free

The moments came from the network, not from the model: $m_{\text{net}}(\theta) = m(\theta) + e(\theta)$.

e does not shrink with the sample

More data makes $\hat{m}$ precise; it does nothing to e . The estimator converges to the minimizer of the criterion built on m_{net} , which is the truth only to the extent that e is small *across the parameter range*.

The practical rule. Report the surrogate's **maximum** consumption error over the parameter box against a reference solution, on the states the model visits, before reporting an estimate. In 07_01 that maximum is $2.1 \cdot 10^{-2}$; in 07_02 it is $3.3 \cdot 10^{-2}$. If it is not small relative to how much the moments move, train longer or widen the state box, and check again.

Chen, Didisheim & Scheidegger (2026), fn. 15, make the same point about their own surrogate.

Finger Exercise: Which Moment Moves With ϱ ?

PAUSE: 5 min

The setting

$\log z_{t+1} = \varrho \log z_t + \sigma \varepsilon_{t+1}$ with $\varepsilon \sim \mathcal{N}(0, 1)$, stationary.

1. Compute $\text{Var}(\log z_t)$ and $\text{Var}(\Delta \log z_t)$ as functions of ϱ and σ .
2. Which of the two rises with ϱ , and which falls?
3. Output is $Y_t = z_t K_t^\alpha$ and capital moves slowly. Which of the four moments on the model frame should respond most to ϱ , and in which direction? Check against the moment figure of Part II.

Solution: Which Moment Moves With ϱ ?

1. The two variances

Stationarity gives $\text{Var}(\log z) = \sigma^2 / (1 - \varrho^2)$. For the difference,

$$\Delta \log z_t = (\varrho - 1) \log z_{t-1} + \sigma \varepsilon_t, \text{ so } \text{Var}(\Delta \log z) = (1 - \varrho)^2 \frac{\sigma^2}{1 - \varrho^2} + \sigma^2 = \frac{2\sigma^2}{1 + \varrho}.$$

2. Directions

The level variance **rises** without bound as $\varrho \rightarrow 1$; the variance of the difference **falls**, from $2\sigma^2$ at $\varrho = 0$ to σ^2 at $\varrho = 1$. A more persistent process wanders further but changes less from one period to the next.

3. The moments

$\text{Corr}(\log Y_t, \log Y_{t-1})$ inherits the autocorrelation of $\log z$, close to ϱ itself, and rises steeply. $\text{Std}(\Delta \log C)$ follows the falling variance of the difference, dampened by consumption smoothing, so it moves little. The mean investment share depends on the level of capital, which ϱ hardly touches. That is the ordering in the Part II figure.

From Classroom to Production

Brock–Mirman is the mechanism in miniature. Chen, Didisheim & Scheidegger (2026) run it on a high-dimensional option-pricing model, with a network in place of a Fourier-transform pricer.

What the surrogate enables

- ▶ Re-estimate **every day** instead of once per paper
- ▶ Systematic out-of-sample evaluation against reduced-form benchmarks
- ▶ Conditional distributions of option returns

What that produced

- ▶ An option-implied **tail-risk index**, predictive of market crashes
- ▶ A **parameter-instability** measure that lines up with option-market illiquidity
- ▶ A read on the structural-versus-reduced-form trade-off

The pattern worth remembering

None of these are faster versions of old results. They are questions whose *answers require a time series of estimates*, and that is what a surrogate with the parameters as inputs manufactures.

Summary, and What to Run

1. The expensive step in SMM is the model solve inside the loop. A policy network with the parameters as inputs is trained once and removes it.
2. Validate the surrogate against a reference solution across the parameter range before estimating with it; the Euler residual is not that validation.
3. Common random numbers make the simulated moments smooth; then a grid shows the criterion and a standard minimizer finishes it.
4. Identification is a property of the moments. A flat valley in the criterion is either the moments or the surrogate; the reference check tells which, and the moment Jacobian puts a number on it.

Hands-on, day2/code/07_structural_estimation/

07_01_SMM_One_Parameter.ipynb (4 min) and 07_02_SMM_Two_Parameters.ipynb (6 min), teaching presets stored; set `RUN_MODE = "smoke"` in class. Try: change the true ϱ ; drop a moment; shorten the observed sample to $T = 100$ and watch the criterion's minimum move.



Questions?

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Next up, Session 8: *Economics-informed neural networks: foundations.*

Materials:

github.com/sischei/deep_learning_padova_0926